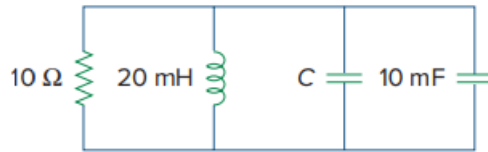


Problem

For the network in Fig. 8.76, what value of C is needed to make the response underdamped with unity neper frequency ($\alpha = 1$)?

Figure 8.76:



Step-by-step solution

Step 1 of 3

Draw the RLC parallel circuit.

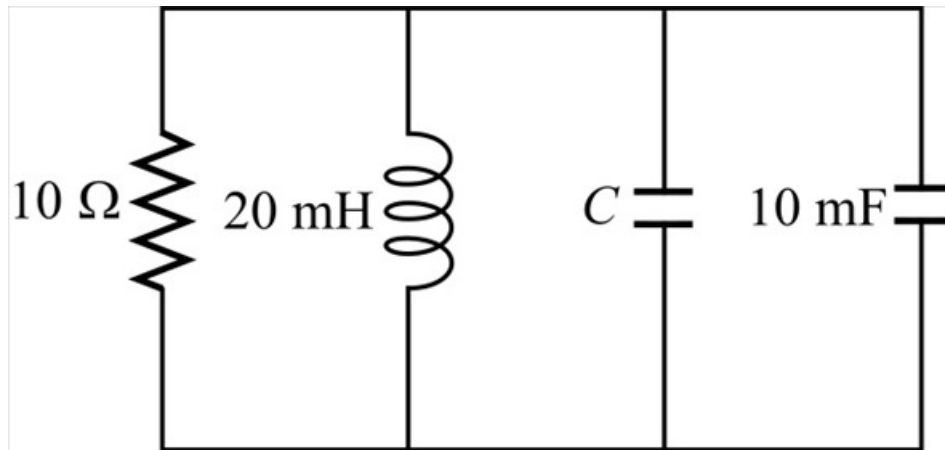


Figure 1

Step 2 of 3

Clearly it is a parallel RLC circuit, given response is underdamped with unity damping factor ($\alpha = 1$).

Write the formula for damping factor in series RLC circuit.

$$\alpha = \frac{1}{2RC_{eq}}$$

$$1 = \frac{1}{(2)(10)(C_{eq})}$$

$$\begin{aligned} C_{eq} &= \frac{1}{(2)(10)} \\ &= 0.05 \text{ F} \end{aligned}$$

Step 3 of 3

Now C_{eq} is the resultant of capacitors C and 10 mF , as they are in parallel resultant is given by:

$$C_{eq} = C + 10 \times 10^{-3}$$

$$0.05 = C + 10 \times 10^{-3}$$

$$C = 0.05 - 0.01$$

$$= 40\text{ mF}$$

Therefore, the value of capacitance C is 40 mF.